

Walmart Sales Analytics Strategy

A comprehensive approach to database management, exploratory data analysis, and dashboard creation for actionable business intelligence.

The Problem Statement

Objective: Transform raw transaction data from multiple Walmart branches into a strategic asset.

🎯 **Goal:** Uncover purchasing trends and evaluate location performance.

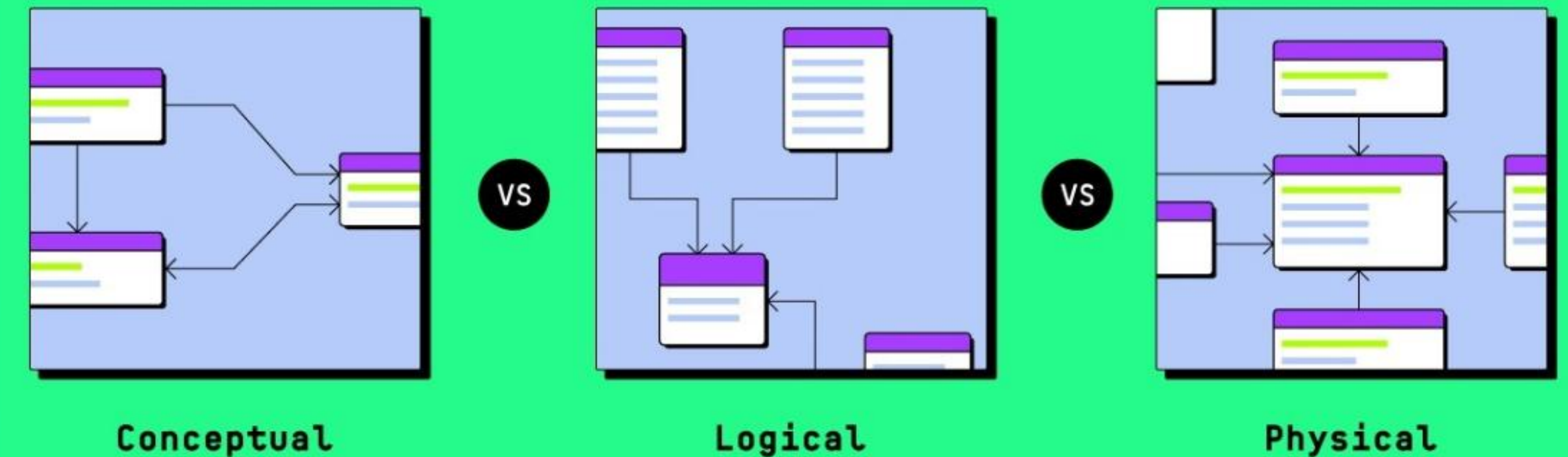
👥 **Target:** Deeply understand consumer behavior across segments.

📈 **Outcome:** Drive data-backed decisions to optimize inventory, tailor marketing, and maximize gross income.



Phase 1: Database Setup & Core SQL

-  **Schema Creation:** Define optimal data types (e.g., VARCHAR for invoice, DECIMAL for financials, TIMESTAMP for dates) and ingest staging data.
-  **Data Validation:** Write queries to check for nulls and validate financial integrity (e.g., ensuring $\text{cogs} + \text{vat} = \text{total}$).
-  **Dimension Modeling:** Normalize the flat CSV into a Star Schema with Dimension tables (Branch, Product, Customer) linked to a central Sales Fact table.



Phase 2: Advanced SQL for BI



Consumer Behavior

Utilize Common Table Expressions (CTEs) and window functions to analyze purchasing patterns by `customer_type` (Prime, Elite, Basic) and gender.



Time-Series Aggregation

Group sales volume and gross income by `time_of_day`, `day_name`, and `month_name` to pinpoint peak shopping hours.

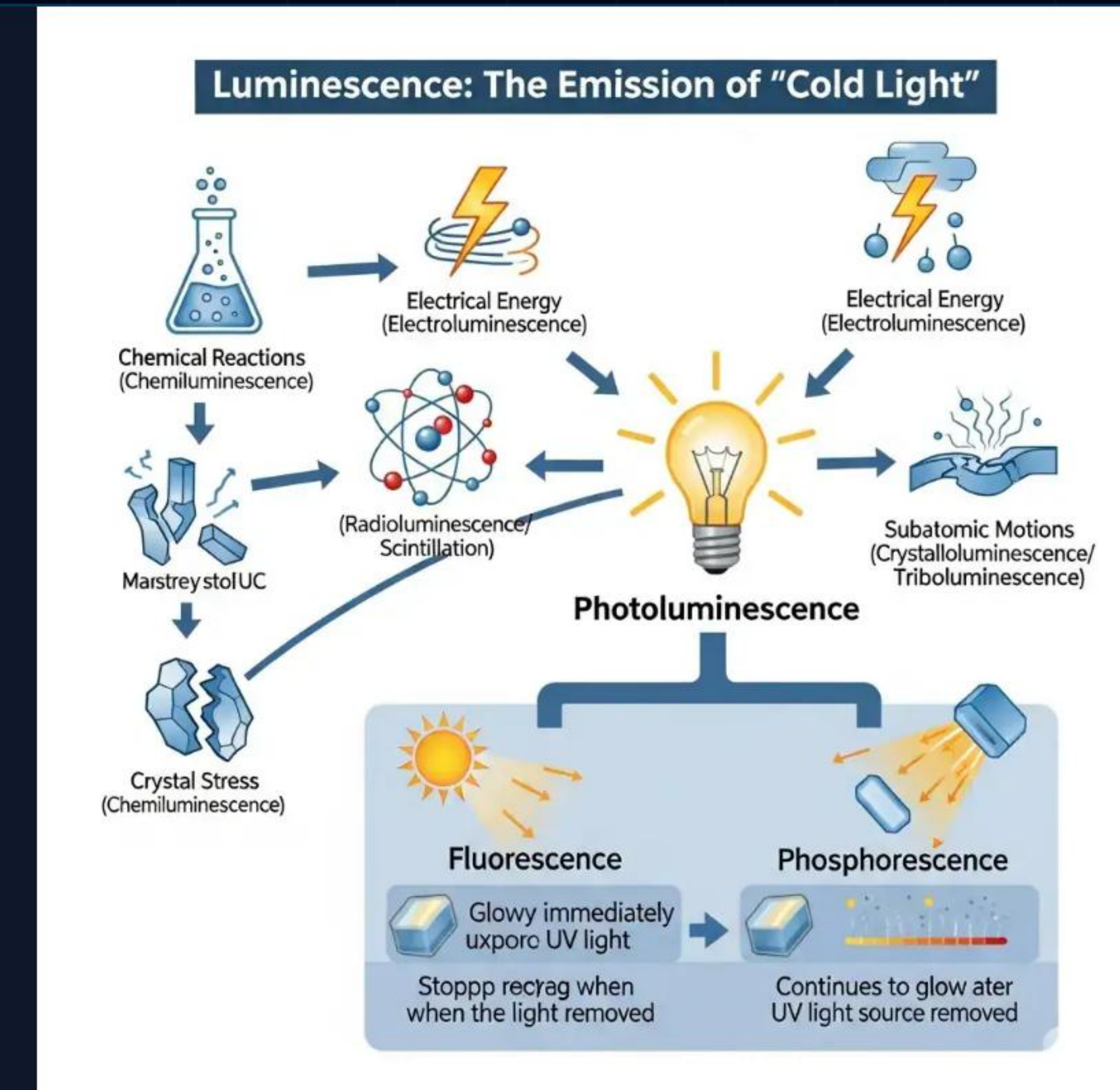


Branch Profiling

Implement ranking queries (`DENSE_RANK()`) to evaluate branches (Banjara Hills, Andheri, CP) based on yield and customer rating.

Phase 3: Exploratory Data Analysis

-  **Distribution Analysis:** Generate histograms for total spend, quantity, and rating to map the spread of spending habits.
-  **Categorical Deep-Dive:** Analyze which product_line moves the highest volume versus which generates the highest gross_income.
- Correlation Checks:** Map relationships between basket size (quantity), COGS, and customer rating to identify satisfaction drivers.



Phase 4: Feature Engineering



Customer Value Tiers

Create a new feature classifying transactions into 'High', 'Medium', or 'Low' Value baskets based on quartiles of the `total` column.



Shopping Shift Matrix

Combine `day_name` and `time_of_day` into a single categorical feature (e.g., "Weekend-Evening") to identify high-traffic windows.



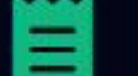
Product Margin Flag

Develop a flag grouping `product_line` items into 'High Margin' vs. 'Low Margin' based on their contribution to overall gross income.

Dashboard Foundation: Core KPIs

AOV

Average Order Value

-  **Total Revenue:** Sum of the total column across selected periods.
-  **AOV:** Average total spend per unique invoice_id.
-  **Gross Income:** Total generated gross_income and average margin percentage.
-  **CSAT Score:** Average rating across branches.

Dashboard Architecture

 **Header:** Title, interactive Date Filter, and Branch/City Dropdown.

 **Scorecards (Row 1):** Big numbers for Revenue, Gross Income, Invoices, CSAT.

 **Primary Charts (Row 2):** Line Chart (Revenue Trends) & Horizontal Bar (Top Products).

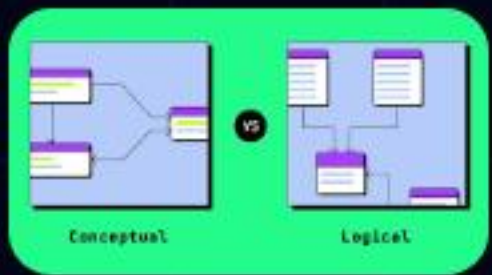
 **Insights (Row 3):** Donut Chart (Customer Type split) & Heatmap (Peak Sales Activity).

Image Sources



<https://www.quantumworkplace.com/hubfs/AI-Generated%20Media/Images/Office%20worker%20looking%20at%20computer%20screen%20of%20charts%20and%20graphs.jpeg>

Source: www.quantumworkplace.com



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